Project Phoenix - Integrated Adaptive Powertrain

System Whitepaper v1.0

Project Phoenix

Document status: Phase 1 simulation complete - no hardware gate cleared

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Classification: External distribution permitted with simulation disclaimer

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Disclaimer: This document reports simulation-validated engineering projections. No road-test, EPA/WLTP certification, or homologation claims are made. Prefix all external statements with ?simulation shows...? Measured bench prototypes are the Seed-phase deliverable.

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1. Executive Summary

Project Phoenix is an integrated series-hybrid powertrain combining:

1. ATPE - crankless three-tier free-piston linear generator
2. PCMRITMS - phase-coordinated multi-rotor inertial torque buffer
3. 20 kWh LFP battery - low-frequency energy store and urban EV range
4. Unified controller - nested slow/fast loops that keep the engine off transients

Physics-based computer simulation stress-tested this architecture across **thirteen validation studies**, six vehicle bodies, temperatures from -10 degC to +40 degC, regulatory-style drive cycles, full payload, cold starts, plug-in charging, 250,000 km ageing, comparison against a conventional 2.0 L turbo, and single-cartridge fault tolerance.

Core conclusion (simulation): a small, cool-running, lifetime-lasting battery plus an inertial sprint reserve plus an efficient generator can deliver strong performance with competitive lifetime cost and carbon - if hardware matches the model assumptions.

What has not been done: building and measuring real cartridges, rotors, or a mule vehicle.

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2. System Architecture

2.1 Energy flow

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PETROL
  ?
  ?
ATPE (3-tier free-piston ring)     ? slow loop: 10-100 ms
  ?
  ?
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DC bus (400-800 V)
  ?
  ??? PCMRITMS (inertial buffer)  ? fast loop: 1-5 ms
  ??? LFP battery (20 kWh)
  ?
  ?
Traction motor(s) -> Wheels
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2.2 Complementary roles

ATPE weakness	PCMRITMS remedy
Tier switching gap (~200-400 ms)	Buffer covers seamlessly
Irregular free-piston ripple	Rotor smoothing
Slow generation setpoint	Millisecond inertial response

PCMRITMS weakness	ATPE remedy
118 kJ depletes in ~1.3 s at full discharge	Engine refills during sustained load
Needs primary energy source	Efficient continuous generation
Round-trip losses	ATPE efficiency advantage offsets

2.3 Reference sizing - AWD SUV (Phase 1)

Subsystem	Parameter	Value
ATPE stack	Peak electrical	230 kW (8 cartridges, 4/2/2 mix)
ATPE stack	Continuous	~100 kW
PCMRITMS	Stored energy	118 kJ
PCMRITMS	Continuous discharge	90 kW
PCMRITMS	Brief burst	140 kW (~1.3 s energy-limited)
PCMRITMS	Peak torque boost	+34.9% (242.8 vs 180 N·m)
Battery	Capacity	20 kWh LFP
Battery	Max discharge (default)	120 kW
Battery	Right-sized discharge (SUV)	90 kW
Battery	Max charge	80 kW
Battery	Charge-sustaining SoC	55%
Traction	Peak power	222 kW
Traction	Peak torque	>=350 N·m
Vehicle	0-100 km/h target	<8 s
Vehicle	Top speed	180 km/h

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3. Unified Control

Three nested loops with strict authority:

Loop A - Mode (100-500 ms): electric-only vs charge-sustaining from filtered demand and SoC floor (25%).

Loop B - ATPE setpoint (10-100 ms): five-second low-pass filtered load plus SoC correction -> smallest sufficient tier set.

Loop C - Buffer arbitration (1-5 ms):

1. PCMRITMS supplies or absorbs mismatch first.
2. Battery covers remainder.
3. Any residual is recorded as capability shortfall - energy is never silently invented.

Surplus generation refills PCMRITMS first, then charges the battery.

Power balance (every timestep): generator + battery + buffer = demand + losses.

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4. Validation Studies - Results (simulation)

Study	Question	Headline result
A - Learned controller	Can control learn from data?	Matches rule-based with fair charge-sustaining correction
B - Battery thermal	Heat and ageing?	+1 degC in normal driving; 0 replacements over vehicle life
C - Sensitivity	Which inputs matter?	Aerodynamics > weight at highway; no single magic factor
D - Economics	True cost and CO??	Battery embodied CO? = 4.7% of SUV lifecycle
E - Smart flywheel timing	Cleverer sprint use?	No benefit - energy-limited, not power-limited
F - Right-sizing	Oversized parts?	SUV battery can drop 120 -> 90 kW discharge
G - Uncertainty	Error bars?	5th-95th percentile bands on all headlines
H - Climate	-10 degC to +40 degC?	12-23% seasonal fuel swing; no thermal derate
I - Regulatory cycles	WLTP/EPA reconstructions?	Rankings match custom cycles
J - Cold start	Cold morning?	Hurts long trips only; urban stays electric
K - Payload	Full load?	More fuel; hill climb still passes
L - Plug-in	Grid charging?	CO? win only on clean grid
M - Ageing	250,000 km?	~23% EV range loss; modest fuel creep
N - ICE benchmark	vs 2.0 L turbo?	28% less fuel (AWD SUV, mixed)

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5. Key Performance Data (simulation)

5.1 AWD SUV headlines

Metric	Value
Highway fuel (charge-sustaining)	4.46 L/100 km
Mixed-cycle fuel	2.35 L/100 km
vs 2.0 L turbo (mixed)	-28% fuel
Mixed-cycle CO?	54 g/km (ICE: 75 g/km)
Lifecycle CO? (cradle-to-grave)	102 g/km
Running cost	EUR 0.072/km [EUR 0.066-0.091]
Battery replacements (250k km)	0
WLTP-class fuel	4.61 L/100 km (ICE: 7.30)
Seasonal fuel swing	11.6%
Cold-start fuel penalty	+19.3%
Full-payload fuel penalty	+19.2%

PHEV CO? (clean grid)	12 g/km
Fuel drift after 250k km	+44.4%
EV range loss after 250k km	-23.3%

5.2 Fleet - mixed cycle (simulation)

Body	Fuel (L/100 km)	Cost (EUR /km)	CO? (g/km)	WLTP (L/100 km)	Season swing
Hatchback	0.55	0.039	44	3.60	22.7%
Sedan	0.64	0.040	47	3.91	22.3%
Crossover	1.11	0.048	60	4.87	20.2%
AWD SUV	2.61	0.072	102	6.07	11.6%
Van / MPV	2.95	0.077	112	6.59	16.6%
Pickup	3.86	0.092	137	7.83	14.8%

5.3 ATPE vs conventional 2.0 L turbo - full comparison (L/100 km)

Body	Urban	Highway	Tow+grade	Mixed	WLTP-class
AWD SUV	0.00 / 0.00	4.46 / 7.67	6.91 / 11.92	2.35 / 3.26	4.61 / 7.30
Sedan	0.00 / 0.00	0.83 / 1.42	4.64 / 9.21	0.40 / 0.64	1.64 / 2.70
Hatchback	0.00 / 0.00	0.69 / 1.19	4.09 / 8.47	0.34 / 0.56	1.18 / 1.92
Crossover	0.00 / 0.00	1.50 / 2.43	5.58 / 10.37	1.01 / 1.46	3.03 / 4.91
Pickup	0.00 / 0.00	6.17 / 9.68	8.58 / 13.62	4.23 / 7.01	6.81 / 10.75
Van / MPV	0.00 / 0.00	5.01 / 8.29	7.39 / 12.42	2.77 / 5.12	5.38 / 8.55

Format: ATPE / conventional ICE. Urban figures are near-zero when battery is charged.

5.4 Fault tolerance

18 of 18 scenarios PASS - one cartridge offline in any tier, all six bodies pass 9/9 performance targets (acceleration, top speed, 20% gradeability, efficiency).

5.5 Virtual single-cartridge bench

48-cell matrix (4 speeds x 4 loads x 3 tiers): 100% pass rate (simulation).

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6. Development Gate Status

Gate	Milestone	Simulation	Hardware
1	Single cartridge bench	48-cell matrix complete	Not started
2	Stable free-piston	Timing model	Not started
3	Linear generator	Parametric model	Not started
4	Multi-cartridge ring	X4-X16 layout study	Not started
5	Vehicle integration	Complete (6 bodies)	Not started
6	AI optimisation	Supervisory brain + ECU (software)	Not started
7	CAD / packaging	Concept renders	Not started

Software position: vehicle integration complete; Gate 6 advanced in software (not certified).

Hardware position: no gate cleared.

Reference vehicle figures use the Phase-1 4/2/2 eight-cartridge stack (~230 kW). A separate production-ring study freezes a 4/6/2 twelve-cartridge layout (~313 kW) - different layer; do not equate the two when quoting fuel economy.

7. Conclusions (simulation)

Supported by modelling:

- Architecture is internally energy-consistent.
- Small LFP pack runs cool, lasts the vehicle life, low embodied carbon (4.7% of lifecycle).
- Capability comes from stored energy and battery power, not peak kW alone.
- Default battery is over-specced - right-sizing saves cost and weight.
- Smart flywheel timing was tested and rejected - honest negative result.
- One cylinder can fail; all bodies still pass performance targets.

Not validated externally:

- Combustion efficiency on a real rig
- Rotor round-trip efficiency and burst containment
- NVH, packaging, homologation
- Official regulatory test traces (reconstructions used)

8. What This Document Does Not Claim

Limit	Implication
No physical prototypes	Efficiency, noise, weight are modelled
Combustion is simplified in default model	Detailed physics used only for bench matrix
WLTP/EPA are reconstructions	Not official copyrighted traces
AI controller is experimental	Not ASIL-certified production software
PCMRITMS 75-85% round-trip	Optimistic until bench measured
Patents not yet filed	File before wide public disclosure

9. Roadmap to Measured Credibility

Step	Months	Deliverable	Model falsified if...
Single ATPE cartridge rig	0-12	Pressure, stroke, kW data	BTE ? 40% at sweet spot
Single PCMRITMS rotor bench	6-18	Torque pulse, round-trip ?	? ? 70% or no +35% boost
Integrated electrical bench	18-24	Shared 800 V bus	Bus instability under spike
Mule vehicle	30-42	Dyno + on-road fuel	Highway fuel ? 6 L/100 km
Emissions + durability	42-54	Euro 7 / 300k km equiv	Regulatory or wear failure

Seed funding (illustrative): $8-15M over 18 months - two bench prototypes with measured data.

10. One-Paragraph Summary

Project Phoenix pairs a crankless three-tier linear generator with a phase-coordinated inertial torque buffer and a small lifetime battery on a unified 800 V DC bus. Thirteen simulation studies across six vehicle types support the conclusion that this architecture can match strong performance with approximately 28% less fuel than a conventional 2.0 L turbo on the same SUV, zero battery replacements over the vehicle life, and **fault tolerance with one cylinder offline**. Simulation shows these outcomes; measured prototypes are the required next step.

Companion documents: ATPE Whitepaper v1.0; PCMRITMS Whitepaper v1.0.